CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Seventh Semester of B. Pharm. Examination
University Theory Examination March/April 2018
PH417 Pharmaceutical Analysis -III

Date: 28.03.2018, Wednesday Time:01.30 p.m to 04.30 p.m Maximum Marks: 80

Instructions:

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat labeled sketches wherever necessary.**

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. The region of electromagnetic spectrum for nuclear magnetic resonance is
[A] Microwave
[B] Radio frequency
[C] Infrared
[D] UV-rays

2. The transition zone for Raman spectra is
[A] Between vibrational and rotational levels
[B] Between electronic levels
[C] Between magnetic levels of nuclei
[D] Between magnetic levels of unpaired electrons

3. Between magnetic levels of unpaired electrons
[A] only the mass of the ions
[B] only the charge of the ions
[C] both the mass and charge of the ions
[D] neither the charge nor mass of the ions

4. ____________ is used to control intensity of light in UV-Visible Spectrophotometer.
[A] Lens
[B] Slit
[C] Mirro
[D] Monochromator

5. ____________ solution is used for calibration of wavelength accuracy of UV-Visible Spectrophotometer
[A] Holmium perchlorate
[B] Potassium dichromate
[C] Potassium chloride
[D] Polystyrene

6. The base peak in a mass spectrum is:
[A] the lowest mass peak.
[B] the peak corresponding to the parent ion.
[C] the highest mass peak.
[D] the peak set to 100% relative intensity.

7. The wavelength of x-rays is measured in
[A] cm
[B] angstrom

[C] micron
[D] nanometer

8. Applications of X-Ray crystallography includes
[A] polymer characterisation
[B] crystal structure
[C] crystallinity of a compound
[D] All of the above

9. In Raman Spectroscopy, Light ____________ is studied.
[A] Absorbed
[B] Transmitted
[C] Scattered
[D] Diffracted

10. Which of the following statement is not true for Raman Spectroscopy?
[A] Raman Spectroscopy can be used in aqueous solutions
[B] vibrations inactive in IR spectroscopy may be seen in Raman spectroscopy.
[C] There is destruction to the sample in Raman Spectroscopy
[D] Raman Spectroscopy needs relative short time

11. Emission of an x-ray photon occurs
[A] Whenever a fast moving electron impinges on the atom, it knocks out the electron from one of the inner shells.
[B] Whenever a fast moving ion impinges on the atom, it knocks out the electron from one of the middle shells.
[C] Whenever a fast moving electron impinges on the atom, it knocks out the electron from one of the outer shells.
[D] Whenever a fast moving electron impinges on the atom, it knocks out the ion from one of the inner shells.

12. ICH stands for
[A] International Council for Harmony
[B] International Committee for Harmonization
[C] Indian Council on Harmonization
[D] International Council on Harmonization

13. Which of the following is NOT a subpart of the GMP guidelines as per FDA regulations?
[A] Buildings and Facilities
[B] Production and Process Controls
[C] Marketing
[D] Returned and Salvaged Drug Products

14. Which of the following is a hard ionisation source used in Mass Spectrometry?
[A] Electron Impact Ionisation
[B] Chemical Ionisation
[C] Field Ionisation
[D] Time of Flight

15. The ion formed by removal of one electron and that occurs mostly at the highest m/z ratio in the mass spectra is called
[A] Isotope ion
[B] Metastable ion
[C] Rearrangement Ion
[D] Molecular Ion

16. Which of the following technique does not require radiation source?
[A] Atomic Absorption Spectroscopy
[B] Atomic Emission Spectroscopy
[C] Atomic Fluorescence Spectroscopy
[D] UV Spectroscopy

17. Which of the following is not suitable solvent for Proton-NMR analysis?
[A] CCl_4
[B] D_2O
[C] DMSO
[D] Chloroform

18. Which of the following statements about tetramethylsilane is incorrect?
[A] it produces a single peak at delta value 10.
[B] it is inert
[C] it is volatile and can be easily distilled off and used again
[D] it is used to provide a reference against which other peaks are measured

19. GMP tends to

[A] Good Material Production
[B] Good Manufacturing Premises
[C] Good Manufacturing Practices
[D] Good Manufacturing and Production

20. When atoms interact with EMR, they show line spectrum because interaction results in only____________.
[A] Electronic transition
[B] Vibrational transition
[C] Rotational transition
[D] All of above

SECTION – II

Q 2	Attempt any **TWO** of the following	
A	What is EMR? Classify.	**05**
B	What is Spectroscopy? Classify them.	**05**
C	Define following terms: i) Wave number, ii) Frequency, iii) Bathochromic shift, iv) Chromophore, v) Auxochrome	**05**
Q 3	Attempt any **TWO** of the following	
A	What is a chemical shift? Describe the factors affecting chemical shift.	**05**
B	Discuss application of NMR spectroscopy.	**05**
C	Explain in detail Sampling techniques in Infrared Spectroscopy.	**05**

SECTION – III

Q 4	Attempt any **FOUR** of the following	
A	Write applications of UV-Visible Spectroscopy	**05**
B	Explain factors affecting intensity of fluorescence.	**05**
C	Write a brief note on Raman Spectroscopy.	**05**
D	Describe instrumentation of Spectrofluorimetry.	**05**
E	Explain general principle of atomic spectroscopy.	**05**
F	Write about principle and design of Hollow Cathode Lamp.	**05**
Q 5	Attempt any **FOUR** of the following	
A	What are thermal methods of analysis? Classify them.	**05**
B	Discuss principle of ANY ONE thermal method.	**05**
C	Explain Brag's Law and derive its equation.	**05**
D	Differentiate QA and QC.	**05**
E	Explain Atomic Spectroscopy and Molecular Spectroscopy.	**05**
F	Write difference between ^{1}H NMR and ^{13}C NMR	**05**